

Ishan Singh

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Summary

Backend / Platform Engineer with 6+ years of experience building distributed systems, multi-tenant SaaS platforms, IAM, real-time data pipelines, and Kubernetes-native backend services at scale.

Technical Skills

Languages: Go, Python, Java, C/C++, SQL

Backend / Infra: Kubernetes, AWS, Kafka, PostgreSQL, ClickHouse, REST, GraphQL

Security / Observability: IAM, SSO (SAML, OIDC), Splunk, CloudWatch

AI / Infra: AWS Bedrock, MCP, LLMs, LangChain, RAG, Qdrant, pgvector

Experience

Cisco

Bengaluru, KA

Software Engineer

Oct. 2024 – Present

- Built tenant-scale global search for Cisco Secure Access using Go, OpenSearch, DynamoDB, and Kafka, indexing 30+ security configuration object types across 50+ searchable fields and millions of cross-object relationships.
- Enabled nested and where-used lookups to trace related objects, policies, and dependencies – targeting 60% lower policy-troubleshooting MTTR and 40% fewer audit misconfigurations.
- Designed anomaly schemas, RCA service APIs (Go), and deployment pipelines (Terraform, Kubernetes) for an internal root-cause analysis platform; shipped MCP servers in Python with Bedrock-powered agents.
- Led implementation of a distributed firewall log ingestion and enrichment pipeline processing 10K events/sec using Go, Kafka, Kinesis, DynamoDB, and S3 with no upstream lag.

Deepfactor, acquired by Cisco

Bengaluru, KA

Senior Software Engineer

Feb. 2023 – Oct. 2024

- Designed and built the IAM layer using Keycloak and Postgres for Deepfactor SaaS beta, supporting multi-tenant authentication and authorization across SaaS and on-prem deployments.
- Implemented generic enterprise SSO support using SAML and OIDC, enabling customer-managed identity provider integrations.
- Designed per-tenant rate limiting at the Kubernetes ingress layer using Istio and custom usage metrics from ClickHouse such as telemetry data ingested and containers observed.
- Built team management and role-scoped access controls for secure multi-tenant collaboration.

Amazon

Bengaluru, KA

Software Development Engineer

Jul. 2021 – Jan. 2023

- Led organization-wide platform migration automation in Java for 50+ teams, reducing a multi-day manual process to a minutes-long workflow.
- Contributed to AIWorkbench (Java), a no-code workflow platform for composing chained workflows across AWS and internal services.
- Built approval workflows that paused execution to validate intermediate ML inference results before continuing downstream processing.
- Built a one-click campaign launcher in Python using ML inference outputs to recommend campaign configurations during large-scale marketplace events.

Optum, UnitedHealth Group

Remote

Software Engineer

Aug. 2020 – Jun. 2021

- Built a service in Java to validate Kafka streams against source databases and generate automated data quality reports.
- Migrated legacy services from Java to Go and added observability using Prometheus, Grafana, and Jaeger.

Mindship

Bengaluru, KA

Software Engineer Intern

Jun. 2019 – Sep. 2019

- Built reusable backend services in Go including a GraphQL-based web chat service and messaging integrations with Twilio and SendGrid.

Projects

Open Agent Policy: Zero-trust access control framework for AI agents with a Python SDK and a policy engine written in Go.

Distributed WAL: Disk-backed distributed write-ahead log in Go with Raft-based replication, leader election, and fault-tolerant recovery.

Lox Interpreter: Tree-walking interpreter for the Lox language in Go, featuring lexical analysis, recursive-descent parsing, and AST evaluation.

Virtual Stylus: Hand-gesture-driven drawing interface built in Python with real-time object detection and classification for touchless HCI.

Education

Zakir Husain College of Engineering & Technology, AMU

Bachelor of Computer Engineering; GPA: 9.12/10

Aligarh, UP

Aug. 2016 – May 2020